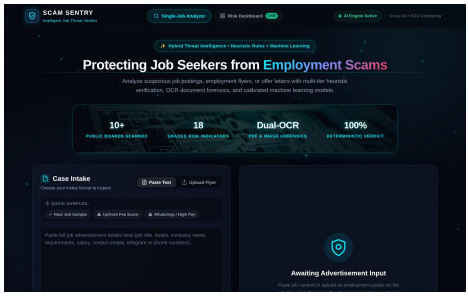
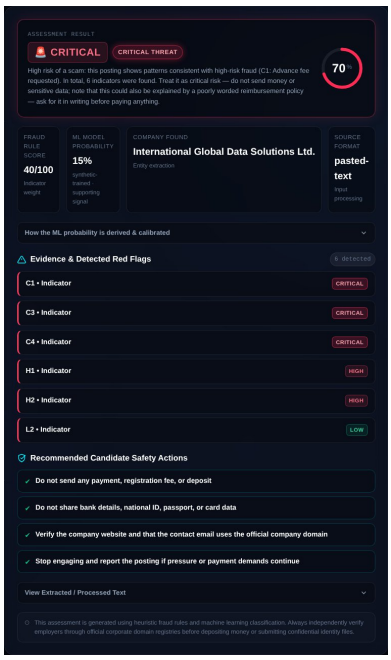
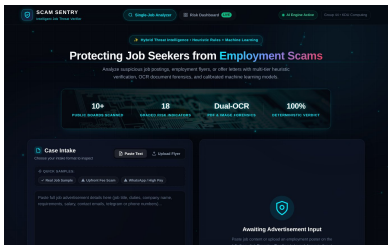
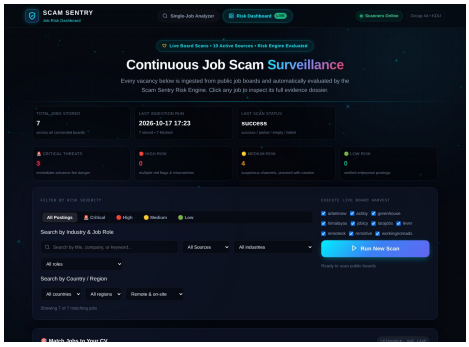
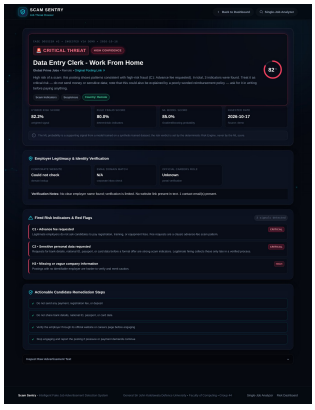
ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

Student Name	Sandun Wickramanayake	**Registration No.**	D/BIT/24/0090
Degree / Intake	Information Technology / 41	**Group No.**	44
Project Title	Scam Sentry: An Intelligent Fake Job Advertisement Detection System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Data pipeline: NLP features, OCR & ML model	Built the original text-processing and detection pipeline in the first commits: text cleaning, TF-IDF vectorisation, hand-crafted fraud-signal features (fraud_features.py) and the model training script (model_training.py) that produced the classifier artifacts. Added the OCR layer (ocr_utils.py) for image and PDF flyers, including the Tesseract fallback path for Windows machines.	Python, scikit-learn TF-IDF, Gradient Boosting pytesseract, pdfplumber	 *Fig 1 - Model metadata served by /api/health*  *Fig 2 - Commit history (git log)*
2. Single-job analyzer interface	Designed and built the main analyzer page: Paste Text and Upload Flyer tabs with drag-and-drop, one-click sample presets (genuine job, upfront-fee scam, WhatsApp/high pay), live character counter and the hero telemetry banner. Wrote the front-end logic (script.js) that calls the analysis API and renders the verdict.	HTML5, CSS3 JavaScript (Fetch API) Flask / Jinja2 templates	 *Fig 3 - Single-Job Analyzer home screen*

3. Verdict report & visual design system	Created the dark glassmorphism theme and cyber colour tokens (style.css, about 4,800 lines changed) and the verdict report: risk banner, circular score gauge, rule / ML score cards, evidence cards with severity badges and the recommended safety actions list. Made the layout responsive.	CSS custom properties SVG gauge Responsive design	 *Fig 4 - Verdict report for a scam posting*
4. Animations & interactive canvas	Implemented the interactive neural-constellation canvas background, the radar animation and the triple staggered laser scan beams. Tuned the 10-second beam cadence and ambient light breathing so the effects stay smooth on lower-end devices.	HTML5 Canvas API CSS keyframe animation Performance tuning	 *Fig 3 - Canvas background and scan hero banner*
5. Risk dashboard, dossier pages & documentation	Upgraded the Risk Dashboard and case-dossier templates (dashboard.html, job_detail.html): KPI tiles, severity filters, grouped job cards and the evidence dossier. Integrated the cyber visual assets, fixed templates and styles to satisfy the CI test suite, and wrote the technical documentation, Canva presentation guide and project slides.	Jinja2 templates Markdown / DOCX / PPTX GitHub Actions (CI)	 *Fig 5 - Risk Dashboard*  *Fig 6 - Case dossier page*

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** 17/10/2026

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.